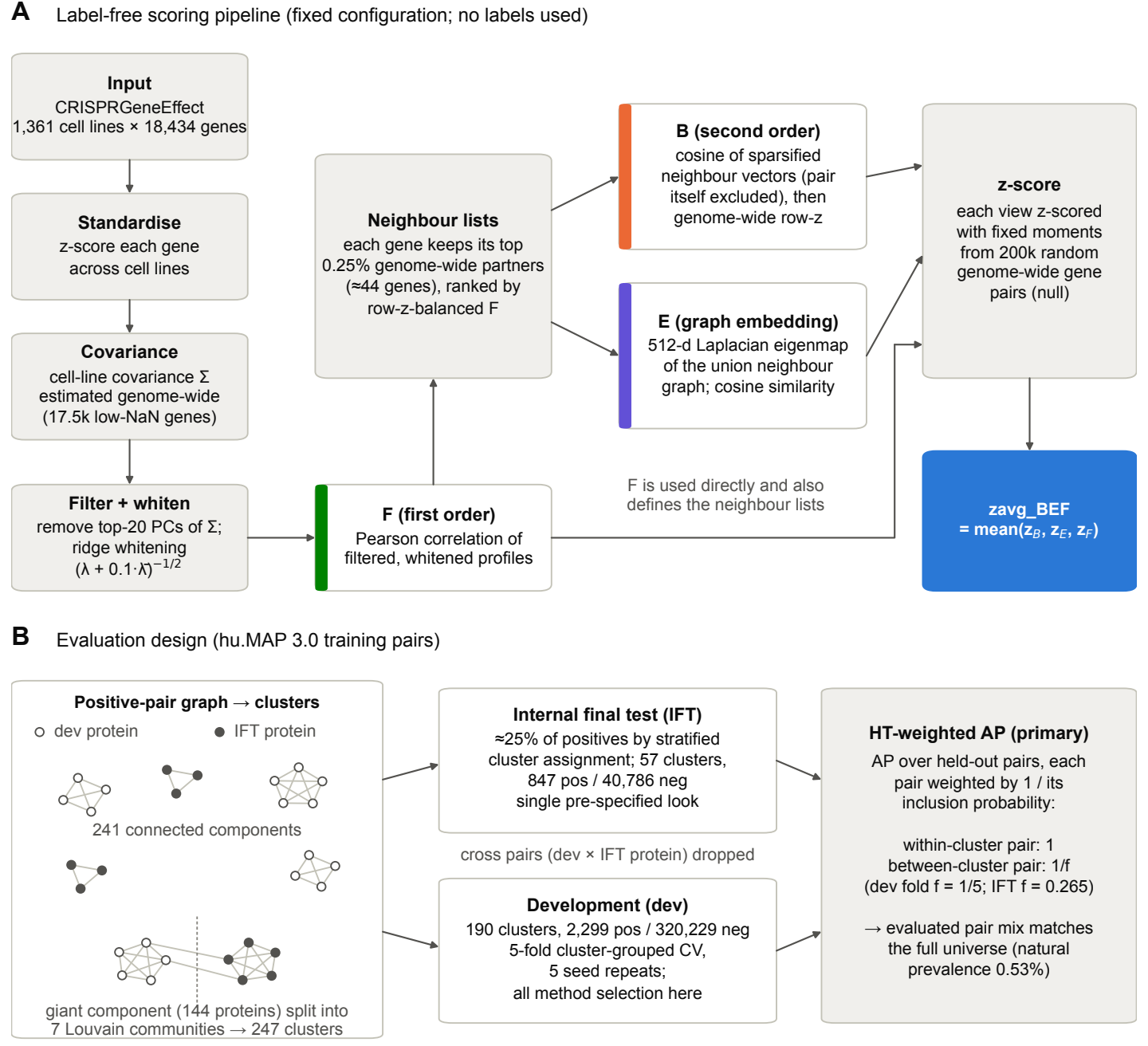

A
Label-free scoring pipeline (fixed configuration; no labels used)
Input
CRISPRGeneEffect
1,361 cell lines × 18,434 genes
Standardise
z-score each gene
across cell lines
Covariance
cell-line covariance Σ
estimated genome-wide
(17.5k low-NaN genes)
Filter + whiten
remove top-20 PCs of Σ;
ridge whitening
(λ + 0.1·λ̄)^−1/2
Neighbour lists
each gene keeps its top
0.25% genome-wide partners
(≈44 genes), ranked by
row-z-balanced F
F (first order)
Pearson correlation of
filtered, whitened profiles
B (second order)
cosine of sparsified
neighbour vectors (pair
itself excluded), then
genome-wide row-z
E (graph embedding)
512-d Laplacian eigenmap
of the union neighbour
graph; cosine similarity
F is used directly and also
defines the neighbour lists
z-score
each view z-scored
with fixed moments
from 200k random
genome-wide gene
pairs (null)
zavg_BEF
= mean(z_B, z_E, z_F)
B
Evaluation design (hu.MAP 3.0 training pairs)
Positive-pair graph → clusters
dev protein
IFT protein
241 connected components
giant component (144 proteins) split into
7 Louvain communities → 247 clusters
Internal final test (IFT)
≈25% of positives by stratified
cluster assignment; 57 clusters,
847 pos / 40,786 neg
single pre-specified look
cross pairs (dev × IFT protein) dropped
Development (dev)
190 clusters, 2,299 pos / 320,229 neg
5-fold cluster-grouped CV,
5 seed repeats;
all method selection here
HT-weighted AP (primary)
AP over held-out pairs, each
pair weighted by 1 / its
inclusion probability:
within-cluster pair: 1
between-cluster pair: 1/f
(dev fold f = 1/5; IFT f = 0.265)
→ evaluated pair mix matches
the full universe (natural
prevalence 0.53%)